

Generative Agents: Interactive Simulacra of Human Behavior

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Figure 1: Generative agents are believable simulacra of human behavior for interactive applications. In this work, we demonstrate generative agents by populating a sandbox environment, reminiscent of *The Sims*, with twenty-five agents. Users can observe and intervene as agents plan their days, share news, form relationships, and coordinate group activities.

ABSTRACT

Believable proxies of human behavior can empower interactive applications ranging from immersive environments to rehearsal spaces for interpersonal communication to prototyping tools. In this paper, we introduce generative agents: computational software agents that simulate believable human behavior. Generative agents wake up, cook breakfast, and head to work; artists paint, while

authors write; they form opinions, notice each other, and initiate conversations; they remember and reflect on days past as they plan the next day. To enable generative agents, we describe an architecture that extends a large language model to store a complete record of the agent's experiences using natural language, synthesize those memories over time into higher-level reflections, and retrieve them dynamically to plan behavior. We instantiate generative agents to populate an interactive sandbox environment inspired by *The Sims*, where end users can interact with a small town of twenty-five agents using natural language. In an evaluation, these generative agents produce believable individual and emergent social behaviors. For example, starting with only a single user-specified notion that one agent wants to throw a Valentine's Day party, the agents autonomously spread invitations to the party over the next two

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days, make new acquaintances, ask each other out on dates to the party, and coordinate to show up for the party together at the right time. We demonstrate through ablation that the components of our agent architecture—observation, planning, and reflection—each contribute critically to the believability of agent behavior. By fusing large language models with computational interactive agents, this work introduces architectural and interaction patterns for enabling believable simulations of human behavior.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools**; • **Computing methodologies** → **Natural language processing**.

KEYWORDS

Human-AI interaction, agents, generative AI, large language models

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1 INTRODUCTION

How might we craft an interactive artificial society that reflects believable human behavior? From sandbox games such as *The Sims* to applications such as cognitive models [23] and virtual environments [10, 59], for over four decades, researchers and practitioners have envisioned computational agents that can serve as believable proxies of human behavior. In these visions, computationally-powered agents act consistently with their past experiences and react believably to their environments. Such simulations of human behavior could populate virtual spaces and communities with realistic social phenomena [27, 80], train people on how to handle rare yet difficult interpersonal situations [44, 52, 94], test social science theories [12, 46], craft model human processors for theory and usability testing [23, 39, 51], power ubiquitous computing applications [31] and social robots [10, 14], and underpin non-playable game characters [59, 85] that can navigate complex human relationships in an open world.

However, the space of human behavior is vast and complex [85, 108]. Despite striking progress in large language models [18] that can simulate human behavior at a single time point [39, 80], fully general agents that ensure long-term coherence would be better suited by architectures that manage constantly-growing memories as new interactions, conflicts, and events arise and fade over time while handling cascading social dynamics that unfold between multiple agents. Success requires an approach that can *retrieve* relevant events and interactions over a long period, *reflect* on those memories to generalize and draw higher-level inferences, and apply that reasoning to create *plans and reactions* that make sense in the moment and in the longer-term arc of the agent's behavior.

In this paper, we introduce *generative agents*—agents that draw on generative models to simulate believable human behavior—and

demonstrate that they produce believable simulacra of both individual and emergent group behavior. Generative agents draw a wide variety of inferences about themselves, other agents, and their environment; they create daily plans that reflect their characteristics and experiences, act out those plans, react, and re-plan when appropriate; they respond when the end user changes their environment or commands them in natural language. For instance, generative agents turn off the stove when they see that their breakfast is burning, wait outside the bathroom if it is occupied, and stop to chat when they meet another agent they want to talk to.¹ A society full of generative agents is marked by emergent social dynamics where new relationships are formed, information diffuses, and coordination arises across agents.

To enable generative agents, we describe an agent architecture that stores, synthesizes, and applies relevant memories to generate believable behavior using a large language model. Our architecture comprises three main components. The first is the *memory stream*, a long-term memory module that records, in natural language, a comprehensive list of the agent's experiences. A memory retrieval model combines relevance, recency, and importance to surface the records needed to inform the agent's moment-to-moment behavior. The second is *reflection*, which synthesizes memories into higher-level inferences over time, enabling the agent to draw conclusions about itself and others to better guide its behavior. The third is *planning*, which translates those conclusions and the current environment into high-level action plans and then recursively into detailed behaviors for action and reaction. These reflections and plans are fed back into the memory stream to influence the agent's future behavior.

This architecture suggests applications in multiple domains, from role-play and social prototyping to virtual worlds and games. In social role-play scenarios (e.g., interview preparation), a user could safely rehearse difficult, conflict-laden conversations. When prototyping social platforms, a designer could go beyond temporary personas to prototype dynamic, complex interactions that unfold over time. For this paper, we focus on the ability to create a small, interactive society of agents inspired by games such as *The Sims*.² By connecting our architecture to the ChatGPT large language model [77], we manifest a society of twenty-five agents in a game environment. End users can observe and interact with these agents. If an end user or developer wanted the town to host an in-game Valentine's Day party, for example, traditional game environments would require scripting tens of characters' behavior manually. We demonstrate that, with generative agents, it is sufficient to simply tell one agent that she wants to throw a party. Despite many potential points of failure—the party planner must remember to invite other agents to the party, attendees must remember the invitation, those who remember must decide to actually show up, and more—our agents succeed. They spread the word about the party and then

¹When referring to generative agents engaging in actions or going to places, this is a shorthand for readability and not a suggestion that they are engaging in human-like agency. The behaviors of our agents, akin to animated Disney characters, aim to create a sense of believability, but they do not imply genuine agency.

²A demonstration of an actual simulation of the generative agent society can be viewed at the following link: https://reverie.herokuapp.com/UIST_Demo/. A public repository for the simulation code is located here: https://github.com/joonspk-research/generative_agents

show up, with one agent even asking another on a date to the party, all from a single user-generated seed suggestion.

We conducted two evaluations of generative agents: a controlled evaluation to test whether the agents produce believable individual behaviors in isolation, and an end-to-end evaluation where the agents interacted with each other in open-ended ways over two days of game time to understand their stability and emergent social behaviors. In the technical evaluation, we leverage a methodological opportunity to evaluate an agent’s knowledge and behavior by “interviewing” it in natural language to probe the agents’ ability to stay in character, remember, plan, react, and reflect accurately. We compared several ablations that limit agents’ access to memory, reflection, and planning. We observe that each of these components is critical to strong performance across these interview tasks. Across the technical and end-to-end evaluation, the most common errors arose when the agent failed to retrieve relevant memories, fabricated embellishments to the agent’s memory, or inherited overly formal speech or behavior from the language model.

In sum, this paper makes the following contributions:

- *Generative agents*, believable simulacra of human behavior that are dynamically conditioned on agents’ changing experiences and environment.
- A novel architecture that makes it possible for generative agents to remember, retrieve, reflect, interact with other agents, and plan through dynamically evolving circumstances. The architecture leverages the powerful prompting capabilities of large language models and supplements those capabilities to support longer-term agent coherence, the ability to manage dynamically evolving memory, and recursively produce higher-level reflections.
- Two evaluations, a controlled evaluation and an end-to-end evaluation, that establish causal effects of the importance of components of the architecture, as well as identify breakdowns arising from, e.g., improper memory retrieval.
- Discussion of the opportunities and ethical and societal risks of generative agents in interactive systems. We argue that these agents should be tuned to mitigate the risk of users forming parasocial relationships, logged to mitigate risks stemming from deepfakes and tailored persuasion, and applied in ways that complement rather than replace human stakeholders in design processes.

2 RELATED WORK

In this section, we reflect on the prior literature in human-AI interaction and situate, within its canon, the agenda of building believable proxies of human behavior. This agenda, once hailed as a north star in the interaction, game, and artificial intelligence communities [10, 59, 85, 86], has remained challenging due to the complexity of human behavior [17, 108]. We synthesize this research to suggest that large language models, though not sufficient by themselves, open up a new angle for creating believable agents when leveraged using the appropriate architecture.

2.1 Human-AI Interaction

Interactive artificial intelligence systems aim to combine human insights and capabilities in computational artifacts that can augment

their users [4, 30]. A long line of work has explored ways to enable users to interactively specify model behavior. For instance, Crayons demonstrated an early vision of interactive machine learning, allowing non-expert users to train classifiers [30]. Further work helped to articulate how end users might describe their classification goals to the system through examples [34] or demonstration [32]. Recent advancements have extended these explorations to deep learning [63] and prompt-based authoring [50, 67, 106].

Meanwhile, a persistent thread of research has advanced the case for language- and agent-based interaction in human-computer interaction. Formative work such as SHRDLU [103] and ELIZA [102] demonstrated the opportunities and the risks associated with natural language interaction with computing systems. As research progressed, it became evident that autonomous agents could offer new metaphors for delegation and interaction [68], but the boundaries of delegation between humans and agents have remained the subject of ongoing debate and refinement [47, 89, 90]. Recently, this technology has reached a level of stability that enables agents to interact via natural language in large and complex online social environments (e.g., [55]). Natural language interaction provides a novel modality that can enhance user abilities in domains such as photo editing [3, 35, 65] and code editing [88].

We convene these threads of work to show that we can now create agents that proxy human behavior for interactive systems, and interact with them using natural language. In doing so, this work reopens the door to examining foundational human-computer interaction questions around cognitive models such as GOMS and Keystroke-Level Model (KLM) [22, 23], around prototyping tools [80], and around ubiquitous computing applications [26, 31, 101].

2.2 Believable Proxies of Human Behavior

Prior literature has described *believability*, or *believable agents*, as a central design and engineering goal. Believable agents are designed to provide an illusion of life and present a facade of realism in the way they appear to make decisions and act on their own volition, similar to the characters in Disney movies [10, 96]. These agents can populate and perceive an open world environment like the one we inhabit [10, 59], and strive to behave in ways that exhibit emergent behaviors grounded in social interactions with users or other agents with the aim of becoming believable proxies of our behavior in hypothetical simulations of individuals and communities [20, 36, 71]. Historically, these agents were developed in the context of intelligent game non-player characters (NPCs) [59, 85]. Creating NPCs with believable behavior, if possible, could enhance player experiences in games and interactive fictions by enabling emergent narratives [8, 16, 49, 93] and social interactions with the agents [109]. However, more importantly, game worlds provide increasingly realistic representations of real-world affordances, and as observed by Laird and van Lent in 2001, these simulated worlds offer accessible testbeds for developers of believable agents to finesse the agents’ cognitive capabilities without worrying about implementing robotics in the real world or creating simulation environments from scratch [59, 85].

A diverse set of approaches to creating believable agents emerged over the past four decades. In implementation, however, these approaches often simplified the environment or dimensions of agent